

ARNAV JOSHI

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Education

Thapar Institute of Engineering and Technology

Aug 2023 – July 2027

B.Tech in Computer Science Engineering

CGPA: 9.57

Ryan International School, Ludhiana

July 2023

Class 12 (Senior Secondary Education)

96.7%

Experience

Samsung R&D Institute, Bengaluru

July 2026 – Dec 2026

Research Intern | Cloud AI/ML

- Building a Retrieval-Augmented Generation (RAG) pipeline for an AI-powered cloud-operations and in-built assistant, enriching natural-language queries with application-specific context.
- Designing an agentic decision layer that lets users accept/reject AI-recommended actions, orchestrating multiple LLM backends into one troubleshooting interface.

IIT Delhi – FE²B Lab

May 2026 – July 2026

AI/ML Intern | Advisor: Dr. Narsing Kumar Jha

- Engineered an AI-based signal classification and adaptive wavelet denoising pipeline to extract low-amplitude acoustic signals from strong background noise, improving signal-to-noise ratio for downstream classification tasks.
- Built MATLAB/Python modal decomposition (SPOD/POD/DMD) and image-processing pipelines on PIV velocity-field data across 6 experimental configurations to characterize acoustic partitioning in multiphase flows.

Technical Skills

Programming Languages: Python, C++, Go (Golang), JavaScript, SQL

Backend: FastAPI, Node.js, REST APIs, Server-Sent Events (SSE), Async Processing, Goroutines & Worker Pools

Databases: PostgreSQL, MongoDB, MySQL, Supabase (Realtime)

Frontend: React.js, Next.js 15, Tailwind CSS, Optimistic UI

AI/ML: PyTorch, TensorFlow, Scikit-learn, Hugging Face Transformers, LangChain, LangGraph, RAG, Vector Databases (Qdrant), OpenCV, NLTK, LLM APIs, MLflow, DVC

DevOps & Tools: Docker, Docker Compose, Kubernetes, AWS EC2, Git, GitHub Actions, CI/CD

Projects

Failure Injection & Observability Engine — *Go, Goroutines, Docker, Distributed Systems*

GitHub

- Built a Go fault-injection backend simulating 5 real-world failure modes (latency, timeout, error, flaky) via goroutine worker pools, with structured JSON logging for full request observability.
- Developed a concurrent log-analysis engine computing P95/P99 latencies and diagnosing root causes across 5 detection rules; shrank the deployment image from 300MB to 20MB (93% reduction) via multi-stage Docker builds.

SynthSales – Agentic AI Sales CRM — *Python, FastAPI, Next.js, PostgreSQL, TypeScript*

GitHub | Live

- Architected an 8-agent pipeline (research, scoring, contact-finding, verification, outreach, follow-up, scheduling, reply classification) with status-driven transitions, 4-way concurrent workers, and snapshot-based rollback for re-runs.
- Engineered a fault-tolerant layer with 3-way LLM failover (Gemini/Groq/OpenRouter) and search-backend failover behind a circuit breaker, plus an email-verification pipeline skipping paid calls via free DNS/MX checks and caching.

Sentinel-RAG – Guarded Enterprise RAG Assistant | *FastAPI, LangGraph, Docker, Qdrant*

GitHub

- Orchestrated a 4-layer guardrail pipeline as a LangGraph state machine behind FastAPI, with structural rail-detection and typed error handling routing every request through injection, topic, and grounding checks in 0.3s.
- Packaged the full stack (LangGraph + local embeddings + reranker) into a 1.36GB non-root Docker image with baked model weights and a fully pinned 149-package dependency closure for reproducible deployment.

Achievements

- Awarded Merit Scholarship for three consecutive years at Thapar Institute of Engineering and Technology, ranking among the top students of the Computer Science branch.
- Achieved a Top 1500 global ranking in the Google Big Code Competition and was selected as a National Semi-Finalist in Flipkart GRiD 8.0.